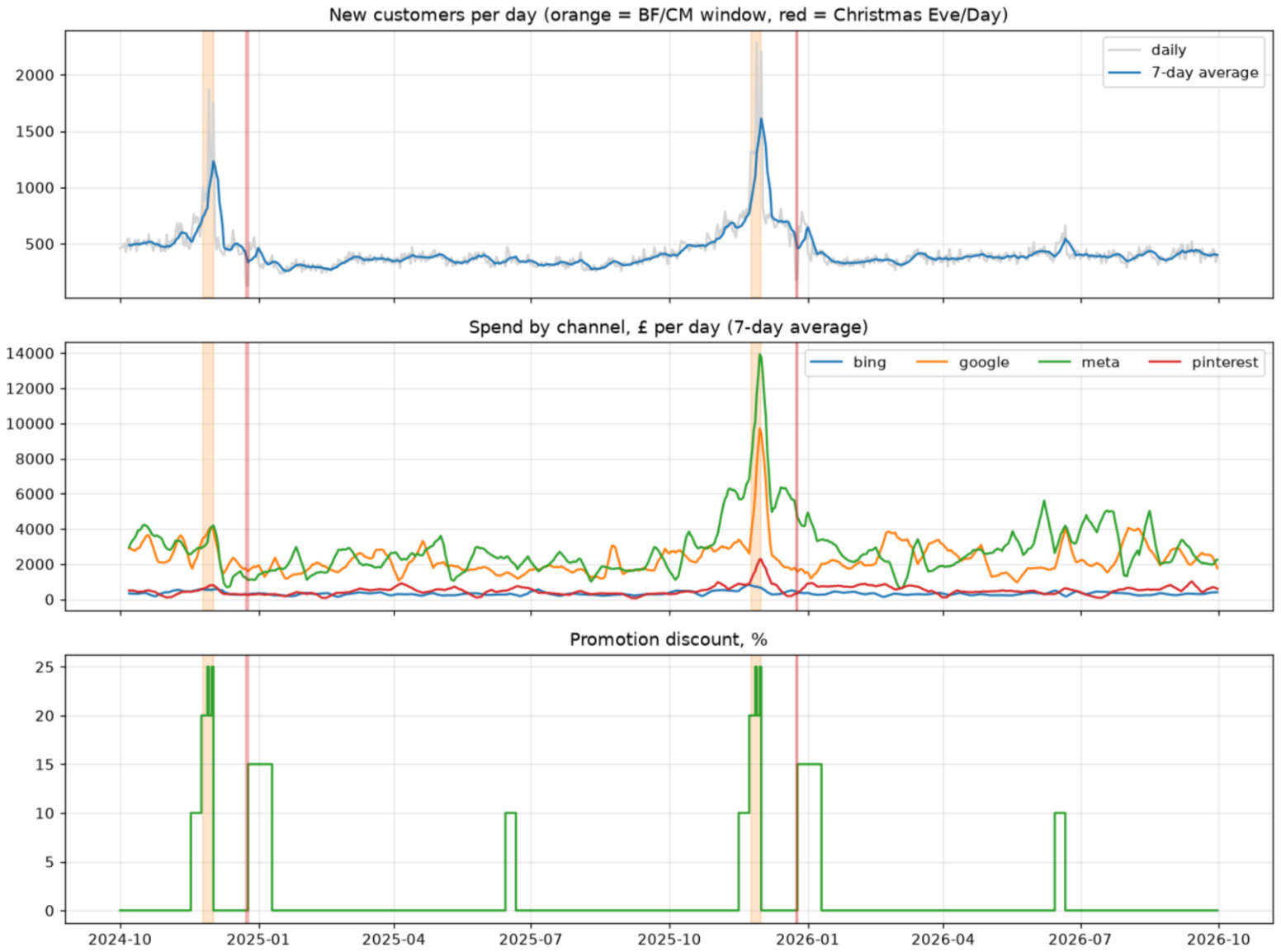
MARKETING MIX MODEL

Where to add budget to win new customers under £40

Marketing mix model, daily data October 2024 to September 2026

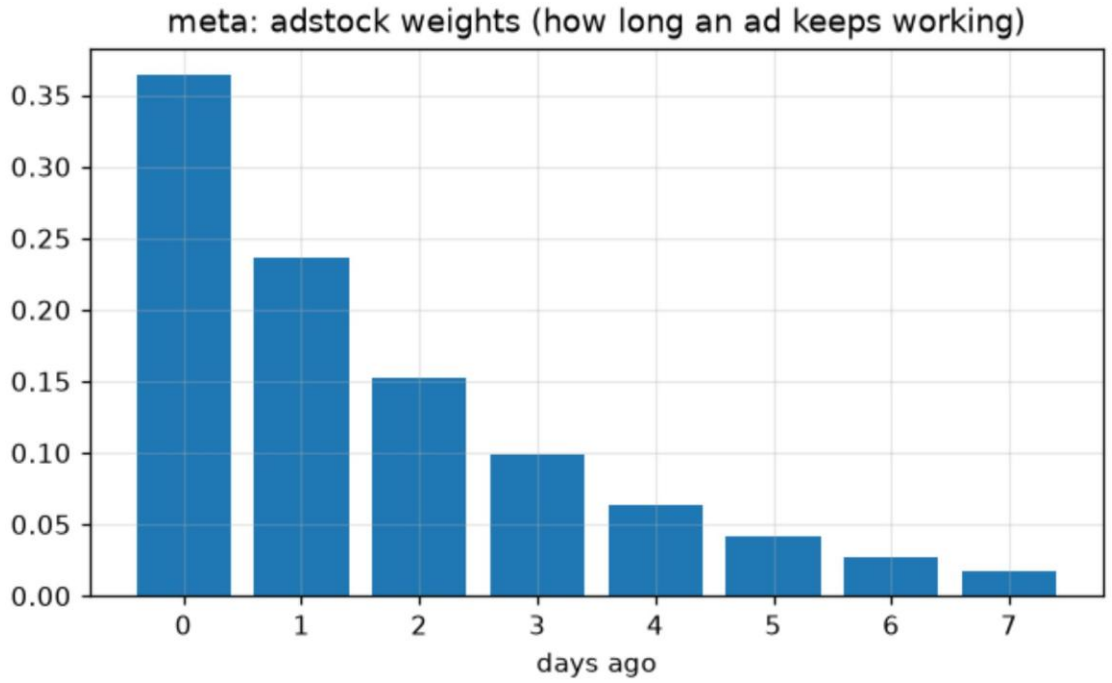
The question is which channels to scale, hold or reduce.

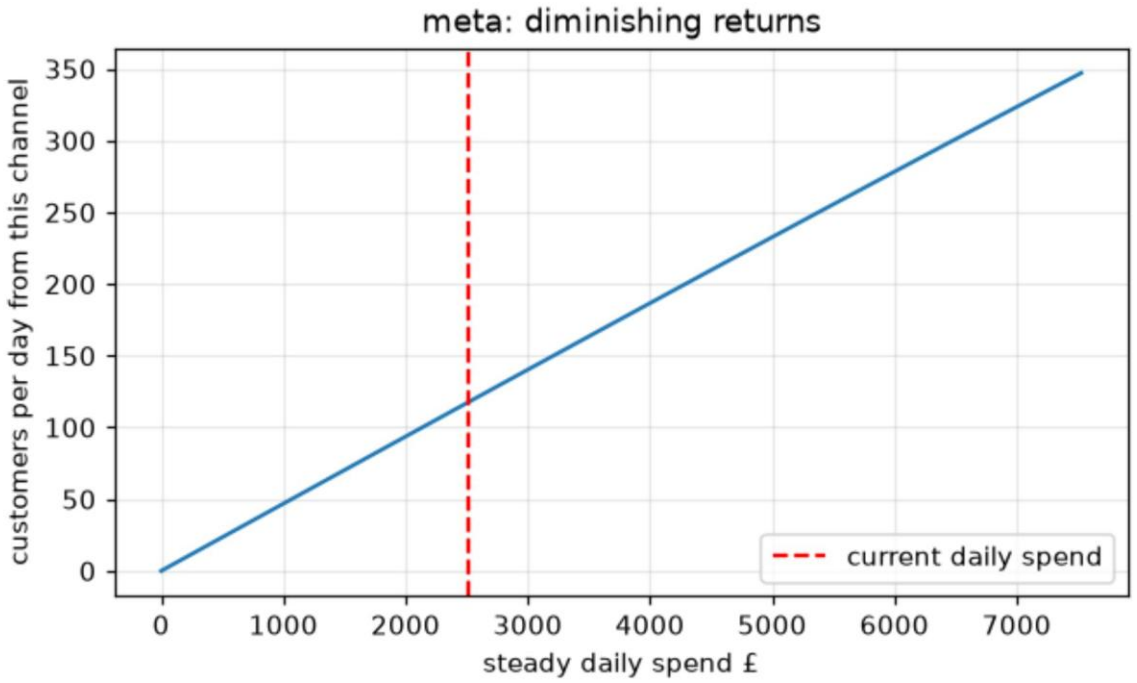


MODEL MECHANICS

How the model works

- Predicts daily new customers from spend in Bing, Google, Meta and Pinterest.
- Separates baseline, seasonality, promotions and Black Friday / Cyber Monday.
- Adstock: Meta’s ads keep working for days; half the effect is gone after about 1.6 days.
- Saturation: Meta’s response is almost a straight line up to 2.5× today’s spend.
- Each estimate is a range, not a single number.



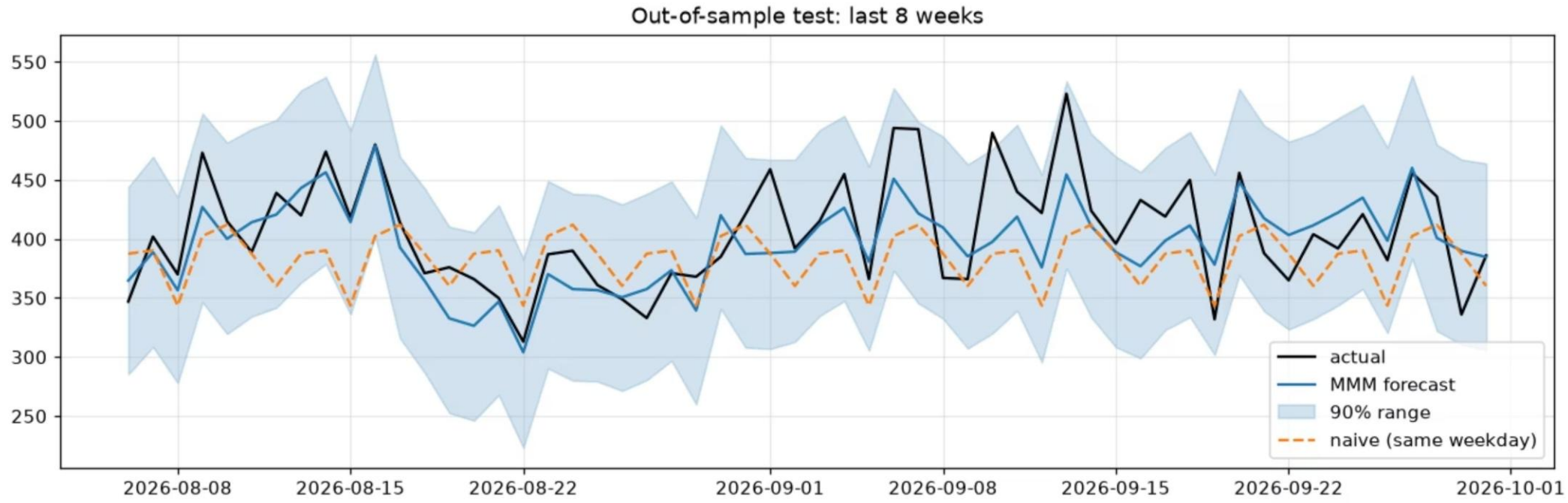


VALIDATION

Can we trust the model?

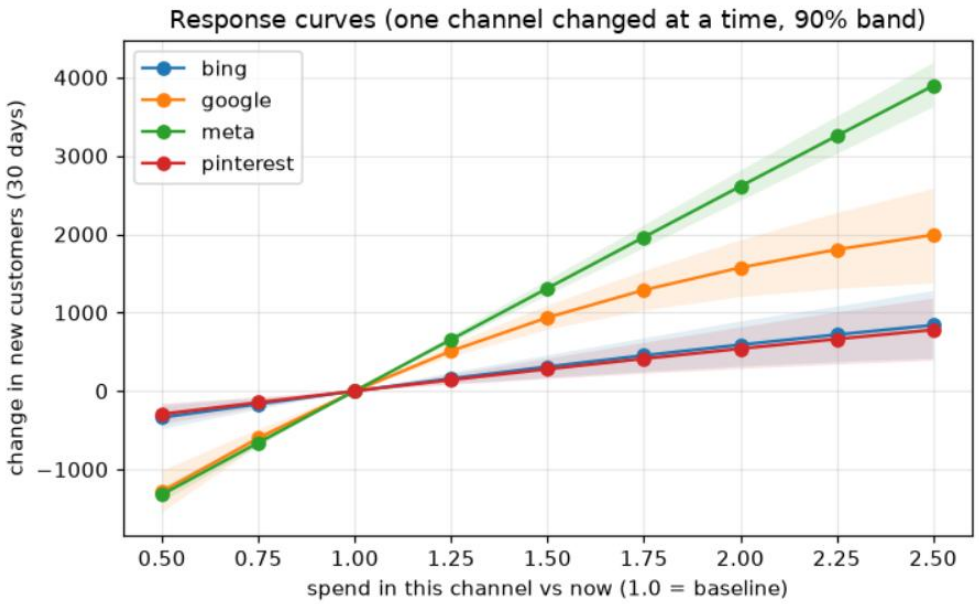
- Data checks: 10 of 10 passed. No gaps, duplicates or unit problems found.
- The sampler converged: max r-hat 1.005, no divergences.
- On the last 8 weeks held out from training, model average error 6.2%; simple forecasts scored 9.1–9.5%.
- The model beat the simple forecasts and regression with no media channels.
- Limit: ranges cover parameter uncertainty only, not the risk that the model structure is wrong.

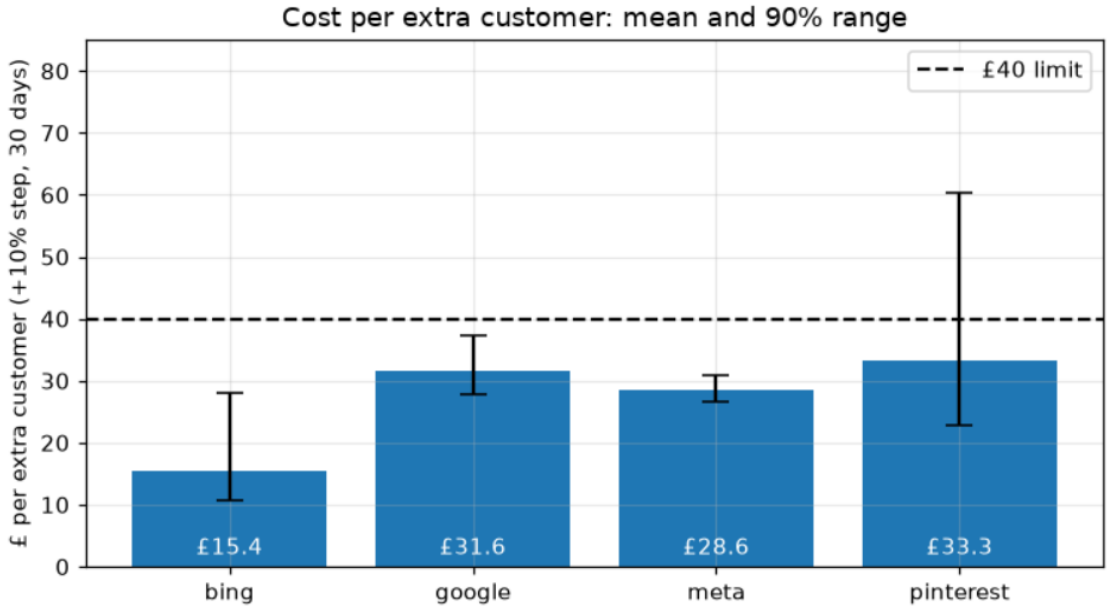
Metric	Model	Same weekday, last 4 weeks	Flat 28-day average	NO media channels
MAE (customers/day)	25.7	38.6	40.4	57.1
MAPE	6.2%	9.1%	9.5%	13.7%
RMSE	32.8	47.9	51.9	69.8

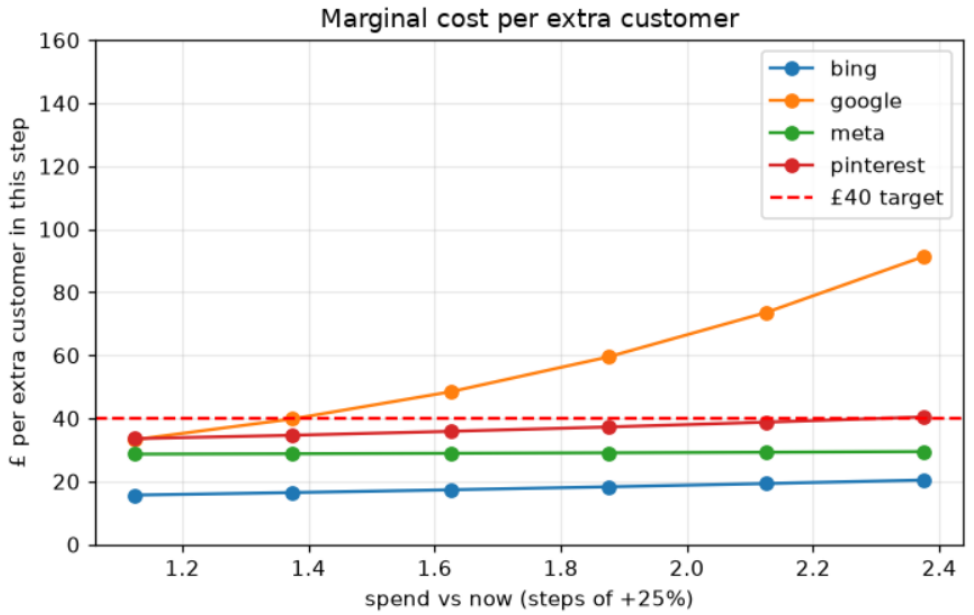


One extra customer costs £15–£33 today; only Pinterest's range crosses £40

- Cost of one extra customer (+10% step, 90% range): Bing £15 (11–28), Meta £29 (27–31), Google £32 (28–37), Pinterest £33 (23–60).
- Meta is the steepest curve and stays near £29 up to 2.5× today's spend; Bing stays under £21.
- Google bends first: reaches £40 at about 1.4 today's spend
- Pinterest is the least certain: 71% chance of staying below £40, and its range reaches £60.





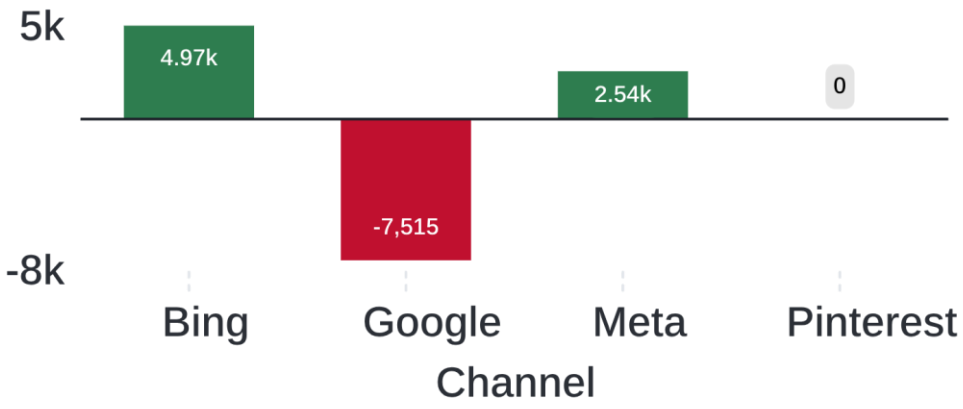


SCENARIO PLANNING

Move £7.5k over 30 days: Bing up, Google down

Same total budget. Expected gain: about +145 new customers in 30 days (about +1% on the baseline of 12,558).

Change in spend over 30 days (£)



Cost of one extra new customer (£)

Channel	Multiplier on today's spend	Now	After the move
Bing	1.50x (cap)	15.4	17.0
Google	0.89x	31.1	29.0
Meta	1.03x	28.6	28.6
Pinterest	1.00x (unchanged)	33.2	33.2

- **Expected gain: +145 new customers over 30 days, 90% range −10 to +296.** The range includes zero, so this is a small, uncertain gain.
- **Bing is capped at 1.5x by our choice.** The data covers spend up to about £1,070 a day
- **Pinterest stays unchanged** because our models disagree on its cost.

Test a 20% Meta increase before scaling further

Acquiring a single customer costs Meta approximately £29, below the £40 limit, so it is the first place to test more budget.

Meta increase	Extra spend	Extra customers	Cost per extra customer
+10%	£7.5k	263	£28.6
+20% (test)	**£15.1k**	**526**	**£28.6**
+40%	£30.1k	1,050	£28.7
+80%	£60.2k	2,093	£28.8

Measure weekly with control regions kept at the old budget → review after 7 days → decide after 30 days

PLAN

- Test: Meta +20% for 30 days, £75.3k → £90.3k (+£15.1k)
- Expected: about 526 extra customers (range 488 to 566)
- Success: cost per extra customer at or below £40; reduce if above £40 for two weeks

LIMITS

- The range shows uncertainty in the estimate, not whether the model itself is right
- Only two Black Fridays in the data
- Correlation is not causation; the test will confirm the effect

Model estimate. The test will confirm or correct it.